

Cartan current tails, adapted near budgets, and temporal packet reduction

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1 Purpose and scope

R-080 left three analytic inputs between the exact R-079 complete packet and the controlled-shell one-use theorem: FAR, NEAR, and the full progressive/revisit extension. This note does not declare any of those three inputs complete. It replaces each broad label by a sharper theorem or obstruction and proves every step that does not require the missing root-resolved production estimate.

The new positive statements are:

1. the production current splits exactly into a spatially quadratic polynomial channel and one heat-averaged nonlinear Cartan channel;
2. the polynomial channel vanishes identically at sufficiently separated output shells;
3. every band-limited deterministic Cartan current has a genuine relative-gap tail of order N^{-2s} for $0 < s < 1$, with the exact critical budget $X^{1/2}Y^{1/2}$;
4. predictable fixed-coefficient Gaussian injection reduces to a triangular half-derivative coefficient norm;
5. an explicitly factorised first-order NEAR martingale response has the required vector-valued Cameron–Martin and terminal-sixth-moment budgets, and therefore its conditional gain ledger is exact on the stochastic-input side;
6. every bounded simple progressive time packet admits strict-past covariance factorisation, and the complete R-079 algebra extends to overlapping temporal packets.

The new negative statements are equally load-bearing. Summing the deterministic FAR estimate separately over probability roots repeats a critical global budget. Subcritical $H^{1/2-}$ coefficient control does not bound the triangular injection norm. A complete nonlinear coefficient innovation does not factor through $D_j = d_j A^*$ alone; its exact Jensen defect can be nonzero when $D_j = 0$. An absolute control–control pair-high extraction is supercritical for the entire candidate NEAR gain range. Finally, the R-075 one-shot graph is not dense in the progressive class even at one scalar mode and in terminal L^2 .

All statements are at finite cutoff, fixed positive production floor, common even regulator, exact products, and the hash-pinned A1 production frame. No root-resolved FAR bound, adapted lower-chaos-complete NEAR operator bound, signed control–control NEAR theorem, overlap-stable progressive packet lower bound, controlled-shell one-use estimate, Nelson bound, measure theorem, removal limit, or Sector-A closure is asserted. The claim remains T4.

2 Authority and production current coordinates

Suppress the spatial direction and the positive Q_{II} contraction. For each symmetric production generator S_A , define

$$\rho_e(z) = |z|^2 + e, \quad n_A(z) = z^T S_A z, \quad q_A(z) = \frac{n_A(z)}{\rho_e(z)}, \quad (2.1)$$

and

$$p_A(z) = 2S_A z, \quad r_A(z) = (S_A - q_A(z)\text{I})z, \quad v_A(z) = 2r_A(z). \quad (2.2)$$

The exact production frame is $M_A = [p_A, v_A]$. The R-068 production identities give

$$|q_A(z)| \leq 1, \quad |r_A(z)| \leq |z|, \quad Dq_A(z)[h] = \frac{2r_A(z) \cdot h}{\rho_e(z)}. \quad (2.3)$$

R-079 supplies the complete expectation-level current/feedback identity, R-080 its regular low bounds and far square completion, and R-063 the deterministic translated balanced-jet reconstruction. Those authorities are used only within their declared scopes.

Introduce the quotient vector

$$F_A(z) = q_A(z)z. \quad (2.4)$$

It is odd, vanishes at zero, and is globally Lipschitz uniformly in the positive floor. Indeed

$$DF_A(z)[h] = q_A(z)h + zDq_A(z)[h]$$

and (2.3) gives

$$|DF_A(z)[h]| \leq |h| + 2\frac{|z|^2}{\rho_e(z)}|h| \leq 3|h|. \quad (2.5)$$

For any centered target Gaussian covariance Σ , translation coupling therefore gives

$$\text{Lip}(P_\Sigma F_A) \leq 3, \quad (P_\Sigma F_A)(0) = 0. \quad (2.6)$$

The constants in later generic current estimates may depend on the fixed floor through higher frame derivatives, but (2.5)–(2.6) do not.

3 Exact polynomial–Cartan current split

Theorem 3.1 (production current split). For every spatially differentiable z ,

$$p_A(z)^T Dz = Dn_A(z), \quad (3.1)$$

and

$$v_A(z)^T Dz = Dn_A(z) - q_A(z)D\rho_e(z) = \rho_e(z)Dq_A(z). \quad (3.2)$$

Moreover, for centered target heat,

$$P_\Sigma p_A = p_A, \quad P_\Sigma v_A = p_A - 2P_\Sigma F_A. \quad (3.3)$$

Consequently the two columns of the heat-averaged spatial current are

$$Dn_A(z), \quad Dn_A(z) - 2(P_\Sigma F_A)(z)^T Dz. \quad (3.4)$$

Proof. Symmetry of S_A gives $Dn_A = 2(S_A z)^T Dz = p_A^T Dz$. Since $D\rho_e = 2z^T Dz$ and $v_A = p_A - 2q_A z$, the first equality in (3.2) follows. Differentiating $n_A = q_A \rho_e$ gives the second. The map p_A is linear and the target Gaussian is centered, while $v_A = p_A - 2F_A$; this proves (3.3) and (3.4). $\square$

If z has spatial Fourier support in $|k| \leq c\Lambda$, then $Dn_A(z)$ has support in $|k| \leq 2c\Lambda$. Hence for a sufficiently large fixed dyadic separation C_0 ,

$$\Pi_m Dn_A(z) = 0 \quad (m \geq j + C_0, \Lambda = 2^j). \quad (3.5)$$

Thus R-080's far loss has no genuinely nonlinear contribution from the p_A column or the Dn_A part of the v_A column. The sole load-bearing spatial tail is the heat-averaged quotient-vector, or Cartan, part of the second column. This is an exact production simplification, not a bound for the full S_C sum.

4 Deterministic FAR theorem and its critical ledger

Let $\Pi_{\geq N\Lambda}$ be a smooth high-frequency projector, let $0 < s < 1$, and let z be spatially band-limited to $c\Lambda$. Put $\bar{F}_A = P_\Sigma F_A$ for any centered target covariance. This is the nonlinear Cartan coefficient isolated in (3.4), not the full two-column frame M_A .

Theorem 4.1 (relative-gap current tail). At the fixed floor,

$$\|\Pi_{\geq N\Lambda}[\bar{F}_A(z)^T Dz]\|_2^2 \leq C_{e,s} N^{-2s} \|z\|_6^2 \|Dz\|_3^2 \quad (4.1)$$

and therefore

$$\|\Pi_{\geq N\Lambda}[\bar{F}_A(z)^T Dz]\|_2^2 \leq C_{e,s} N^{-2s} \|z\|_{H^2} \|z\|_6^3. \quad (4.2)$$

Proof. Oddness gives $\bar{F}_A(0) = 0$, and (2.6) gives the uniform Lipschitz ceiling three after target convolution. The fractional product and Nemytskii estimates give

$$\|\bar{F}_A(z)^T Dz\|_{H_2^s} \lesssim \|\bar{F}_A(z)\|_6 \|Dz\|_{W^{s,3}} + \|\bar{F}_A(z)\|_{W^{s,6}} \|Dz\|_3. \quad (4.3)$$

Lipschitz composition and Bernstein bound the right side by $C_{e,s} \Lambda^s \|z\|_6 \|Dz\|_3$. The high-frequency tail estimate $\|\Pi_{\geq N\Lambda} f\|_2 \lesssim (N\Lambda)^{-s} \|f\|_{H_2^s}$ proves (4.1). Finally the three-dimensional interpolation

$$\|Dz\|_3 \lesssim \|z\|_{H^2}^{1/2} \|z\|_6^{1/2} \quad (4.4)$$

gives (4.2). $\square$

With $X = \|z\|_{H^2}^2$ and $Y = \|z\|_6^6$, the right side of (4.2) is

$$C_{e,s} N^{-2s} X^{1/2} Y^{1/2}. \quad (4.5)$$

There is a genuine and arbitrarily high subunit relative-gap exponent in N , but the energy ledger at one probability root is exactly critical. If a root-resolved theorem allowed the factor $2^{-\theta C}$ to remain after the j sum, a sufficiently large fixed C and scalar Young would pay arbitrary $\eta \mathbb{E}X + \zeta \mathbb{E}Y + C_{\eta,\zeta}$. Applying (4.2) independently for every j instead repeats the same global X/Y payment and does not prove R-080's

$$S_C = \mathbb{E} \sum_j \sum_{m \geq j+C} \|\Pi_m d_j \mathcal{J}_j\|_{Q_{\text{II}}}^2. \quad (4.6)$$

The precise sufficient FAR successor is a martingale-root-resolved estimate of the form

$$\sum_j \mathbb{E} \|\Pi_{\geq j+C} d_j \mathcal{J}_j\|_{Q_{\text{II}}}^2 \leq C 2^{-\theta C} \{1 + \mathbb{E}[X^{1/2} Y^{1/2}]\}, \quad \theta > 0, \quad (4.7)$$

with derivative injection, value innovation, and heat compensator retained together. Neither R-050's unshifted enhancement nor R-063's deterministic translation theorem supplies (4.7) for random predictable translations.

5 Fixed-coefficient injection and the half-derivative endpoint

The predictable part of derivative injection can be isolated exactly. Let g_j be a fresh Gaussian shell and let f be independent of g_j . Fourier support gives, for $m \geq j + C$,

$$\Pi_m(fDg_j) = \Pi_m\{(\tilde{\Pi}_m f)Dg_j\}. \quad (5.1)$$

Using the derivative-shell covariance $\mathbb{E}|Dg_j(x)|^2 \lesssim 2^j$ yields

$$\mathbb{E}_{g_j}\|\Pi_m(fDg_j)\|_2^2 \lesssim 2^j\|\tilde{\Pi}_m f\|_2^2. \quad (5.2)$$

If the same coefficient f occurs at every j , summing first in j gives

$$\sum_m \sum_{j \leq m-C} \mathbb{E}\|\Pi_m(fDg_j)\|_2^2 \lesssim 2^{-C}\|f\|_{H^{1/2}}^2. \quad (5.3)$$

For predictable coefficients f_j , the exact sufficient object becomes the triangular norm

$$\mathfrak{C}_C(f_\bullet) = \sum_m \sum_{j \leq m-C} 2^j \mathbb{E}\|\tilde{\Pi}_m f_j\|_2^2. \quad (5.4)$$

For production the natural predictable coefficient is $f_j = (P_{\Sigma_{j+1}:J} F_A)(Z_{j-1})$. The rest of the base increment still contains the same-shell nonlinear difference and the heat compensator, so (5.4) is a sharp subchannel reduction, not an identity for all of S_C .

The exponent $1/2$ in (5.3) is a true endpoint. Take orthogonal dyadic coefficients with

$$\|\Pi_m f^{(J)}\|_2^2 = 2^{-m}, \quad 1 \leq m \leq J. \quad (5.5)$$

For every $\delta > 0$,

$$\sup_J \|f^{(J)}\|_{H^{1/2-\delta}}^2 = \sup_J \sum_{m=1}^J 2^{-2\delta m} < \infty, \quad (5.6)$$

whereas (5.4) with the same coefficient at every root grows as

$$\sum_{m>C} 2^{-m} \sum_{j \leq m-C} 2^j \asymp 2^{-C} J. \quad (5.7)$$

Therefore an $H^{1/2-}$ coefficient theorem alone cannot close FAR. The balanced R-063 subtraction, square-trace cancellation, or a genuine martingale Carleson estimate must create the missing root summation. This is a method no-go, not a counterexample to the complete production packet.

6 NEAR: vector budget and nonlinear factorisation boundary

Let ℓ be the fixed low cutoff and define the high terminal control martingale differences

$$D_j = d_j(A^* - P_\ell A^*), \quad D = (D_j)_j. \quad (6.1)$$

All spatial norms below are taken with values in ℓ_j^2 . Hilbert-valued Doob orthogonality and the control-energy identification give

$$\mathbb{E}\|D\|_{H_x^2(\ell^2)}^2 \lesssim \mathbb{E}X. \quad (6.2)$$

Pointwise scalar Burkholder followed by spatial integration gives, up to the already fixed-low term,

$$\mathbb{E}\|D\|_{L_x^6(\ell^2)}^6 \lesssim \mathbb{E}Y. \quad (6.3)$$

The vector-valued version of the R-079 interpolation is

$$\|D\|_{H_x^{2/5}(\ell^2)} \lesssim \|D\|_{H_x^2(\ell^2)}^{1/5} \|D\|_{L_x^6(\ell^2)}^{4/5}. \quad (6.4)$$

Thus every response branch already factorised linearly through D_j has the exact stochastic input cost

$$X_D^{1/10} Y_D^{2/15}. \quad (6.5)$$

It is essential not to promote (6.5) to the whole nonlinear future coefficient. Such a coefficient is not determined by D_j . Let $\varepsilon_r = \text{sgn } g_r$, where g_j, g_{j+1} are independent fresh Gaussian roots (bounded smooth approximants give the same limit), let ψ_p be a nonzero bounded deterministic carrier in a later shell $p > j + 1$, and take the progressive control

$$A(x) = \lambda \varepsilon_{j+1} (1 + c \varepsilon_j) \psi_p(x) e_1, \quad 0 < c < 1. \quad (6.6)$$

Then

$$d_j A = d_j D A = 0, \quad d_j |A|^2 = 2c \lambda^2 \varepsilon_j |\psi_p|^2. \quad (6.7)$$

This blind spot occurs in the actual production coefficient. With $\alpha = 5/9$ and the positive diagonal current weights c_0, c_1 , the pure-doublet production trace is

$$h(t) := \text{Tr } B(te_1) = 12c_0 t^2 + c_1 \left\{ 12t^2 - \frac{8\alpha t^4}{t^2 + e} + \frac{4\alpha^2 t^6}{(t^2 + e)^2} \right\}. \quad (6.8)$$

Consequently

$$d_j h(A) = \frac{\varepsilon_j}{2} \{h(\lambda(1+c)\psi_p) - h(\lambda(1-c)\psi_p)\} = 24(c_0 + c_1)c\lambda^2 \varepsilon_j \psi_p^2 + O_e(\lambda^4 |\psi_p|^4) \neq 0 \quad (6.9)$$

as a nonzero function in L_x^1 for all sufficiently small nonzero λ . Thus this production trace component is not determined by D_j . The witness does not by itself exclude cancellation after recombination with the complete square-trace-forest packet; rather, no componentwise or complete representation can be inferred from D_j alone without first proving cancellation of this Jensen defect.

There is a canonical nonduplicating split of this blind spot. Put $a_j = P_j A^*$ and, for a deterministic C^1 coefficient H , define its conditional Jensen defect

$$\mathfrak{D}_j^H = P_j H(A^*) - H(a_j). \quad (6.10)$$

For every active root, $D_j = a_j - a_{j-1}$ and exactly

$$d_j H(A^*) = \int_0^1 D H(a_{j-1} + t D_j) D_j dt + (\mathfrak{D}_j^H - \mathfrak{D}_{j-1}^H). \quad (6.11)$$

The first term is the genuine D_j secant. The second is an upper-triangular conditional-variance/all-higher-chaos feedback generated by later roots; it is the nonzero term in (6.9). Production rational coefficients have no finite-chaos truncation, so this Jensen-defect branch must remain complete. Identity (6.11) is for one fixed deterministic or predictable H . If the coefficient family itself changes with the root, the additional heat or projector compensator $[H_j - H_{j-1}](a_{j-1})$ must also be retained.

There is nevertheless a precise conditional first-order route. If an explicitly projected response branch $R_j^{(1)}$ has the exact factorisation $R_j^{(1)} = \mathbb{K}_j^{\text{near}}[D_j]$, a sufficient vector-valued adapted estimate is

$$\|\{\mathbb{K}_j^{\text{near}}[D_j]\}_j\|_{H_x^{-3/5+\gamma}(\ell^2)} \leq R_\gamma \|D\|_{H_x^{2/5}(\ell^2)}, \quad R_\gamma \in \bigcap_{p<\infty} L^p. \quad (6.12)$$

If (6.12) holds for any $\gamma > 0$, pairing it with the R-078 payload at $\beta = 3/5 - \gamma$ gives

$$R_\gamma X^{1/2-\gamma/4} Y^{1/2+\gamma/12}. \quad (6.13)$$

The exact ledger is

$$a = \frac{1}{2} - \frac{\gamma}{4}, \quad b = \frac{1}{2} + \frac{\gamma}{12}, \quad c = 1 - a - b = \frac{\gamma}{6}, \quad p_R = \frac{6}{\gamma}. \quad (6.14)$$

At $\gamma = 1/20$ this is

$$(a, b, c, p_R) = \left(\frac{39}{80}, \frac{121}{240}, \frac{1}{120}, 120 \right). \quad (6.15)$$

The budgets (6.2)–(6.5) and ledger (6.13)–(6.15) are proved for that declared factorised branch, conditional on (6.12). The remaining nonlinear oscillation coordinate

$$\Omega_j(H) = d_j H(A^*) \quad (6.16)$$

must be estimated as a coefficient/complete-packet martingale square function in its own right and kept with the trace and forest. R-063 proves deterministic translated finite-forest bounds, not either adapted extension.

Strict-past measurability does not freeze the coefficient at probability root j . For $p > j$, a bounded centered $\phi(\xi_j)$ gives the admissible later control

$$a_p = K_p[\phi(\xi_j)e_p], \quad d_j a_p = a_p \neq 0. \quad (6.17)$$

Its canonical weight $2^{4p}\mathbb{E}\|d_j a_p\|_2^2$ is independent of $p - j$ after normalization. Hence the hidden future coefficient in R-080 cannot be replaced by an $\mathcal{F}_{j-1}$ -measurable deterministic translate.

7 The control–control pair-high branch must remain signed

A fresh-Gaussian/control high pair may be compatible with (6.12), because the balanced Gaussian jet can in principle create gain without spending another control factor. The pure control–control pair-high route is different. Suppose an absolute estimate exposes one additional comparable control factor with interpolation fraction $\vartheta \in [0, 1]$. Its budget is

$$2^{-\vartheta p} X^{\vartheta/2} Y^{(1-\vartheta)/6}. \quad (7.1)$$

Adding (7.1) to the first-order gain ledger (6.13) changes the Young slack to

$$c_{\text{pair}} = \frac{\gamma}{6} - \frac{\vartheta}{2} - \frac{1-\vartheta}{6} = \frac{\gamma - 1 - 2\vartheta}{6} < 0 \quad (7.2)$$

for every $0 < \gamma < 1/10$ and every $\vartheta \geq 0$. At $\vartheta = 0$ the loss is already $(\gamma - 1)/6$.

Therefore the control–control pair-high branch cannot be closed by exposing and absolutely paying another control factor. It must remain signed with the terminal current square, covariance trace, paid subtraction, and complete R-063 forest. R-080's exact negative root fixture already excludes universal pointwise square–trace positivity. The precise second NEAR successor is thus a production-specific signed control–control balanced-root lemma. Equation (7.2) forbids one architecture; it does not falsify that signed lemma or the complete one-use estimate.

8 Progressive temporal packet factorisation

Let $I_k = (t_{k-1}, t_k]$ be one interval of a finite time partition and let $u_t = u_k$ there, with u_k bounded, cylindrical, and $\mathcal{F}_{t_{k-1}}$ -measurable. Put

$$\Delta C_k = \int_{I_k} J_t J_t^* dt, \quad L_k = \int_{I_k} J_t dt, \quad a_k = L_k u_k. \quad (8.1)$$

Theorem 8.1 (strict-past temporal factorisation). There is a predictable h_k such that

$$a_k = \Delta C_k^{1/2} h_k, \quad \|h_k\|^2 \leq |I_k| \|u_k\|^2. \quad (8.2)$$

Proof. Operator Cauchy–Schwarz gives

$$L_k L_k^* \preceq |I_k| \Delta C_k. \quad (8.3)$$

Douglas factorisation writes $L_k = \Delta C_k^{1/2} R_k$ with $\|R_k\| \leq |I_k|^{1/2}$. Take $h_k = R_k u_k$. $\square$

The Gaussian increment on I_k has covariance ΔC_k and is independent of $\mathcal{F}_{t_{k-1}}$, so every bounded simple progressive packet enters the R-064 strict-past causalisation exactly. The ranges $\text{Ran } \Delta C_k$ for different k may overlap or coincide. That overlap does not damage Hilbert martingale orthogonality, the deterministic covariance split, or the simultaneous complete-coefficient split. Therefore the complete-current and complete-safe-packet algebra of R-079 temporalises verbatim, including the f_k - i_k cross term.

What does not temporalise is just as important. The R-075 whole-shell graph, R-066 cumulative orthogonal-energy calculation, and R-080 separate low and far estimates use mutually orthogonal physical ranges. They cannot be applied packet by packet after the ranges overlap. The complete endpoint must first be recombined, and only then estimated uniformly in time mesh and revisit multiplicity.

9 Exact non-density of the one-shot graph

The progressive gap is not a technical density omission. It is already visible at one scalar cutoff. Let $J_t = 1$ and let W be one Brownian mode. The bounded simple progressive control

$$u_t = \tanh(W_{1/2}) \mathbf{1}_{(1/2, 1]}(t) \quad (9.1)$$

has terminal displacement

$$A(u) = \frac{1}{2} \tanh(W_{1/2}). \quad (9.2)$$

Any regular one-shot displacement in this only physical range is $\mathcal{F}_0$ -measurable. Even if independent auxiliary randomness is placed in $\mathcal{F}_0$, it remains independent of $W_{1/2}$. Since $\mathbb{E}A(u) = 0$, for every such displacement a ,

$$\mathbb{E}|A(u) - a|^2 = \mathbb{E}|A(u)|^2 + \mathbb{E}|a|^2 \geq \mathbb{E}|A(u)|^2 > 0. \quad (9.3)$$

Thus there is no terminal- L^2 approximation, hence no approximation in the stronger R-075 graph containing terminal L^6 .

Refining time makes each temporal packet strict-past but leaves every packet in the same physical covariance range. Orthogonalising the noise coordinates does not make those physical ranges orthogonal while preserving the original filtration and causality. This proves that regular one-shot graph density is not a route to the full progressive theorem.

The exact progressive successor is an overlap-stable complete-packet bound. Writing

$$G_J(\phi) = V_J^{\text{ren}}(\phi) + \epsilon_6 \|\phi\|_6^6,$$

one must prove, uniformly in J , in the time partition, and in revisit multiplicity,

$$\mathbb{E}G_J(X_J + h_J(u)) + \epsilon_v \mathbb{E} \sum_k |I_k| \|u_k\|^2 \geq -C \quad (9.4)$$

for every bounded cylindrical simple progressive control. Only after (9.4) may one use fixed-cutoff admissible-core truncation, auxiliary conditioning, terminal graph continuity, and lower semicontinuity to reach every finite-energy Boue–Dupuis control. Predictable L^2 density alone does not control the terminal L^6 coordinate or an infinity-minus-infinity limit.

10 Exact successor theorem and conditional synthesis

R-081 reduces the immediate frontier to the following three inputs:

- (FAR) the root-resolved complete-current estimate (4.7), or an equivalent signed square–trace–forest theorem;
- (NEAR-G) an exact first-order branch projection followed by the vector-valued adapted estimate (6.12) for one $\gamma > 0$ on the fresh-Gaussian/control response;
- (NEAR-C) a production-specific signed theorem for the nonlinear Jensen-defect oscillations (6.11)/(6.16) and control–control balanced roots, retaining every square, trace, forest, compensator, and paid subtraction;
- (PROG) the overlap-stable bounded-simple complete-packet lower bound (9.4), followed by the fixed-cutoff variational extension.

NEAR-G and NEAR-C are two branches of the single NEAR gate, not two new claim cards. The witness (6.6)–(6.9) forbids merging NEAR-C into a map of D_j alone. FAR and NEAR may eventually be proved by one vector-valued martingale paracomposition theorem, but no such theorem is attributed here.

Once these inputs are available, the exact R-080 conditional arithmetic is unchanged. With

$$\epsilon_6 = 0.15, \quad \epsilon_v = 0.45, \quad p = \frac{11}{10},$$

the sextic margin is $0.27 - 0.06 - 0.15 = 0.06$, the control margin is

$$\frac{1}{2p} - \epsilon_v = \frac{1}{220}, \quad (10.1)$$

and

$$q = \frac{1}{2\epsilon_v} = \frac{10}{9}, \quad q - p = \frac{1}{90}. \quad (10.2)$$

These equations prove budget compatibility only. They do not prove the one-use inequality, the Nelson exponential moment, or a Gibbs measure.

11 Evidence map, failures, and research roadmap

The proof map after R-081 is

$$\begin{array}{c}
 \text{R-079 complete current} \longrightarrow \text{R-080 LOW and far square} \\
 \downarrow \\
 \boxed{\text{polynomial FAR vanishes}} \quad \boxed{\text{Cartan FAR has deterministic gap}} \\
 \boxed{\text{factorised NEAR input budget closes}} \\
 \downarrow \quad \text{FAR + complete NEAR still required} \\
 \text{regular complete packet lower bound} \\
 \downarrow \quad \text{overlap-stable packet bound required} \\
 \text{full progressive one-use} \longrightarrow q = 10/9 \text{ Nelson, conditionally.}
 \end{array} \tag{11.1}$$

The reusable successes and failed routes are:

- **Closed:** exact polynomial/Cartan current split and far disappearance of the polynomial channel.
- **Closed:** deterministic relative-gap current tail for every $0 < s < 1$.
- **Failed route:** summing that deterministic lemma independently over roots; it repeats a critical global budget.
- **Closed reduction:** fixed-coefficient injection is governed by a triangular half-derivative norm.
- **Failed route:** using only $H^{1/2-}$ coefficient regularity; the explicit endpoint sequence grows linearly.
- **Closed:** vector-valued adapted input budget and exact positive-gain ledger for an explicitly factorised first-order NEAR response.
- **Failed route:** representing the whole nonlinear future coefficient through $D_j = d_j A^*$; (6.6)–(6.9) give $D_j = 0$ but a nonzero production coefficient innovation.
- **Failed route:** absolute control–control pair-high harvest; its slack is strictly negative throughout the candidate range.
- **Closed:** bounded-simple temporal covariance factorisation and algebraic temporalisation of the complete R-079 packet.
- **Failed route:** density of the R-075 one-shot graph in general progressive controls, already at terminal L^2 in one mode.

The next proof effort should not retry generic target heat smoothing, a shellwise deterministic FAR sum, coefficient freezing, an absolute extra control factor, or one-shot density. It should construct a single vector-valued martingale paracomposition estimate that retains the complete square–trace–forest block and controls the nonlinear oscillations (6.16), then prove its overlap-stable temporal version.

12 Devil’s-advocate review

1. **Objection:** “The polynomial/Cartan split already proves FAR.” **UPHELD as false.** It removes the polynomial channel. The nonlinear Cartan current, its value innovation, and heat compensator remain in (4.7).

2. **Objection: “The factor 2^{-2sC} can be made small and paid root by root.” UPHELD as false.** Each use costs the same critical $X^{1/2}Y^{1/2}$ budget. A cutoff-uniform proof must retain the gain after the martingale-root sum.
3. **Objection: “The Lipschitz coefficient is automatically $H^{1/2}$.” UPHELD as false.** The production field is naturally at the $1/2$ – endpoint. Equations (5.5)–(5.7) show that every fixed $H^{1/2-\delta}$ bound is compatible with linear triangular divergence.
4. **Objection: “Equation (5.4) is the whole FAR current.” UPHELD as false.** It covers the predictable fixed-coefficient injection subchannel. The same-shell nonlinear difference and heat compensator must remain in the complete root-resolved theorem.
5. **Objection: “The NEAR adapted operator is now proved.” UPHELD as false.** Equations (6.2)–(6.5) prove the D_j input budget and (6.13)–(6.15) its conditional ledger only after an exact first-order factorisation. Equations (6.6)–(6.9) prove that the complete nonlinear coefficient does not factor through D_j ; (6.11), (6.12), (6.16), and the signed control–control branch remain hypotheses.
6. **Objection: “Pair-high failure disproves NEAR.” DISMISSED.** Equation (7.2) refutes only absolute extraction of an extra control factor. A signed production cancellation retaining the complete companions remains possible.
7. **Objection: “Temporal factorisation proves the progressive lower bound.” UPHELD as false.** It proves strict-past packet representation and complete algebra. Range overlap invalidates the separate R-080 estimates, so (9.4) remains a new analytic theorem.
8. **Objection: “Auxiliary randomness repairs one-shot density.” UPHELD as false.** Independent auxiliary randomness has zero cross term with the centered witness (9.2), so the positive distance (9.3) persists.
9. **Objection: “The exact margins (10.1)–(10.2) prove Nelson.” UPHELD as an overclaim.** They become operative only after FAR, NEAR, and PROG close.

13 Reproduction and independent audit

The primary executable checks the production quotient identities, the uniform Lipschitz ceiling, target-heat contraction, exact Fourier support, fixed-coefficient injection covariance, the half-derivative endpoint sequence, FAR and NEAR rational ledgers, finite-tree Doob/Burkholder fixtures, temporal Cauchy–Douglas factorisation, a complete R-079 packet identity with nonzero injected/future cross term, the progressive non-density witness, and the conditional Nelson arithmetic. The non-importing executable uses an alternate Pauli generator, finite differences, random Fourier data, an independent endpoint sequence, a three-level martingale, a matrix Douglas factorisation, and Gauss–Hermite quadrature.

Run from the repository root:

```
python codes/foundations/
a13_classii_cartan_tail_adapted_near_temporal_reduction_verify.py
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The child contracts are primary 60/60 and non-importing independent 47/47. The integrated verifier pins R-063, R-075, R-079, and R-080; checks source, note, PDF, manifest, run, result, negative-result, gate, roadmap, task, claim, exploration, and honesty surfaces; and fails closed if any FAR, NEAR, progressive, one-use, Nelson, measure, or Sector-A flag is promoted.

Result footer

Result ID: A13-CLASSII-CARTAN-TAIL-ADAPTED-NEAR-TEMPORAL-
REDUCTION

Precise statement: The production current has an exact quadratic-polynomial/heat-Cartan split. The polynomial far channel vanishes; the Cartan current has a genuine deterministic relative-gap bound but still needs a martingale-root-resolved sum. An explicitly factorised first-order NEAR input has the exact vector Doob-Burkholder budget, but the complete nonlinear coefficient has an additional Jensen-defect innovation. Absolute control-control pair-high payment is supercritical. Bounded-simple progressive packets factor causally and preserve complete algebra, but the R-075 one-shot graph is not progressive-dense.

Scope: Finite cutoff; fixed positive floor; common even regulator; exact products; hash-pinned production frame; constants cutoff-uniform only where stated.

Dependencies: A1; R-050; R-063; R-064; R-066; R-070; R-075; R-078; R-079; R-080.

Evidence grade: ANALYTIC, EXACT, EXECUTED.

Reproduction command: python codes/foundations/
a13_classii_cartan_tail_adapted_near_temporal_
reduction_verify.py

Expected output: Primary 60/60; independent 47/47; manifest-pinned integrated and aggregate PASS; exit zero.

Falsification gate: Failure of the current split, relative-gap theorem, endpoint sequence, vector budget, nonlinear-D_j witness, Jensen split, pair-high ledger, temporal factorisation, non-density witness, evidence hash, PDF contract, or honesty flag.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: No root-resolved FAR theorem, complete NEAR theorem, overlap-stable progressive lower bound, controlled-shell one-use, Nelson bound, measure theorem, removal limit, or Sector-A closure is proved.

Next required action: Prove the complete root-resolved FAR estimate and adapted/signed NEAR theorem, then prove the overlap-stable bounded-simple packet lower bound and its variational extension.